

Internet Of Things

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Module 3 : The Core IoT Functional Stack

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Books Referred

- David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things", 1st Edition, Pearson Education (Cisco Press Indian Reprint). (ISBN: 9789386873743)

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- IoT networks are built around the concept of **“things,” or smart objects.**
- These objects are “smart” because they use a combination of contextual information and configured goals to perform actions.
 - ? These actions can be **self-contained** (that is, the smart object does not rely on external systems for its actions);
 - ? however, in most cases, **the “thing” interacts with an external system to report information that the smart object collects, to exchange with other objects, or to interact with a management platform.**
 - The management platform can be used to process data collected from the smart object and also guide the behavior of the smart object.

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- The components of an IoT network:
 - ? Things Layer
 - ? Communications network layer:
 - Access network sublayer
 - Gateways and backhaul network sublayer
 - Network transport sublayer
 - IoT network management sublayer
 - ? Application and analytics layer

- **“Things” layer:** At this layer, the physical devices need to fit the constraints of the environment in which they are deployed while still being able to provide the information needed.

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- **Communications network layer:** When smart objects are not self-contained, they need to communicate with an external system. In many cases, this communication uses a wireless technology.
 - ? This layer has four sublayers:
 - ? **Access network sublayer:** The last mile of the IoT network is the access network. This is typically made up of wireless technologies such as 802.11ah, 802.15.4g, and LoRa. The sensors connected to the access network may also be wired.
 - ? **Gateways and backhaul network sublayer:** A common communication system organizes multiple smart objects in a given area around a common gateway. The gateway communicates directly with the smart objects. The role of the gateway is to forward the collected information through a longer-range medium (called the backhaul) to a headend central station where the information is processed. This information exchange is a Layer 7 (application) function, which is the reason this object is called a gateway. On IP networks, this gateway also forwards packets from one IP network to another, and it therefore acts as a router.
 - ? **Network transport sublayer:** For communication to be successful, network and transport layer protocols such as IP and UDP must be implemented to support the variety of devices to connect and media to use.
 - ? **IoT network management sublayer:** Additional protocols must be in place to allow the headend applications to exchange data with the sensors. Examples include CoAP and MQTT

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- **Application and analytics layer:**
- At the upper layer, an application needs to process the collected data, not only to control the smart objects when necessary, but to make intelligent decision based on the information collected and, in turn, instruct the “things” or other systems to adapt to the analyzed conditions and change their behaviors or parameters.

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Layer 1: Things: Sensors and Actuators

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- From an architectural standpoint, the variety of smart object types, shapes, and needs drive the variety of IoT protocols and architectures.
- The architectural classification could be:
 - ? Battery-powered or power-connected
 - ? Mobile or static
 - ? Low or high reporting frequency
 - ? Simple or rich data
 - ? Report range
 - ? Object density per cell

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Layer 1: Things: Sensors and Actuators

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□ **Battery-powered or power-connected:**

- ? Based on whether the object carries its own energy supply or receives continuous power from an external power source.
- ? Battery-powered things can be moved more easily than line-powered objects. However, batteries limit the lifetime and amount of energy that the object is allowed to consume, thus driving transmission range and frequency.

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Layer 1: Things: Sensors and Actuators

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□ **Mobile or static:**

- ? “thing” should move or always stay at the same location.
- ? A sensor may be mobile because it is moved from one object to another (for example, a viscosity sensor moved from batch to batch in a chemical plant) or because it is attached to a moving object (for example, a location sensor on moving goods in a warehouse or factory floor).
- ? The frequency of the movement may also vary, from occasional to permanent.
- ? The range of mobility (from a few inches to miles away) often drives the possible power source.

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Layer 1: Things: Sensors and Actuators

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- ❑ **Low or high reporting frequency:**
 - ? how often the object should report monitored parameters.
 - ? A rust sensor may report values once a month.
 - ? A motion sensor may report acceleration several hundred times per second.
 - Higher frequencies drive higher energy consumption, which may create constraints on the possible power source (and therefore the object mobility) and the transmission range.

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Layer 1: Things: Sensors and Actuators

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□ **Simple or rich data:**

- ? the quantity of data exchanged at each report cycle.
- ? A humidity sensor in a field may report a simple daily index value (on a binary scale from 0 to 255), while an engine sensor may report hundreds of parameters, from temperature to pressure, gas velocity, compression speed, carbon index, and many others.
- ? **Richer data typically drives higher power consumption.**
- ? This classification is often combined with the previous to determine the object data throughput (low throughput to high throughput).
- ? A medium-throughput object may send simple data at rather high frequency (in which case the flow structure looks continuous), or may send rich data at rather low frequency (in which case the flow structure looks bursty).

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Layer 1: Things: Sensors and Actuators

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□ **Report range:**

? the distance at which the gateway is located.

- For example, for your fitness band to communicate with your phone, it needs to be located a few meters away at most.
- The assumption is that your phone needs to be at visual distance for you to consult the reported data on the phone screen.
- If the phone is far away, you typically do not use it, and reporting data from the band to the phone is not necessary.
- A moisture sensor in the asphalt of a road may need to communicate with its reader several hundred meters or even kilometers away.

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Layer 1: Things: Sensors and Actuators

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- ❑ **Object density per cell:**
- ❑ the number of smart objects (with a similar need to communicate) over a given area, connected to the same gateway.
 - ? An oil pipeline may utilize a single sensor at key locations every few miles.
 - ? telescope at the Whipple Observatory deploy hundreds, and sometimes thousands, of mirrors over a small area, each with multiple gyroscopes, gravity, and vibration sensors.

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Layer 1: Things: Sensors and Actuators

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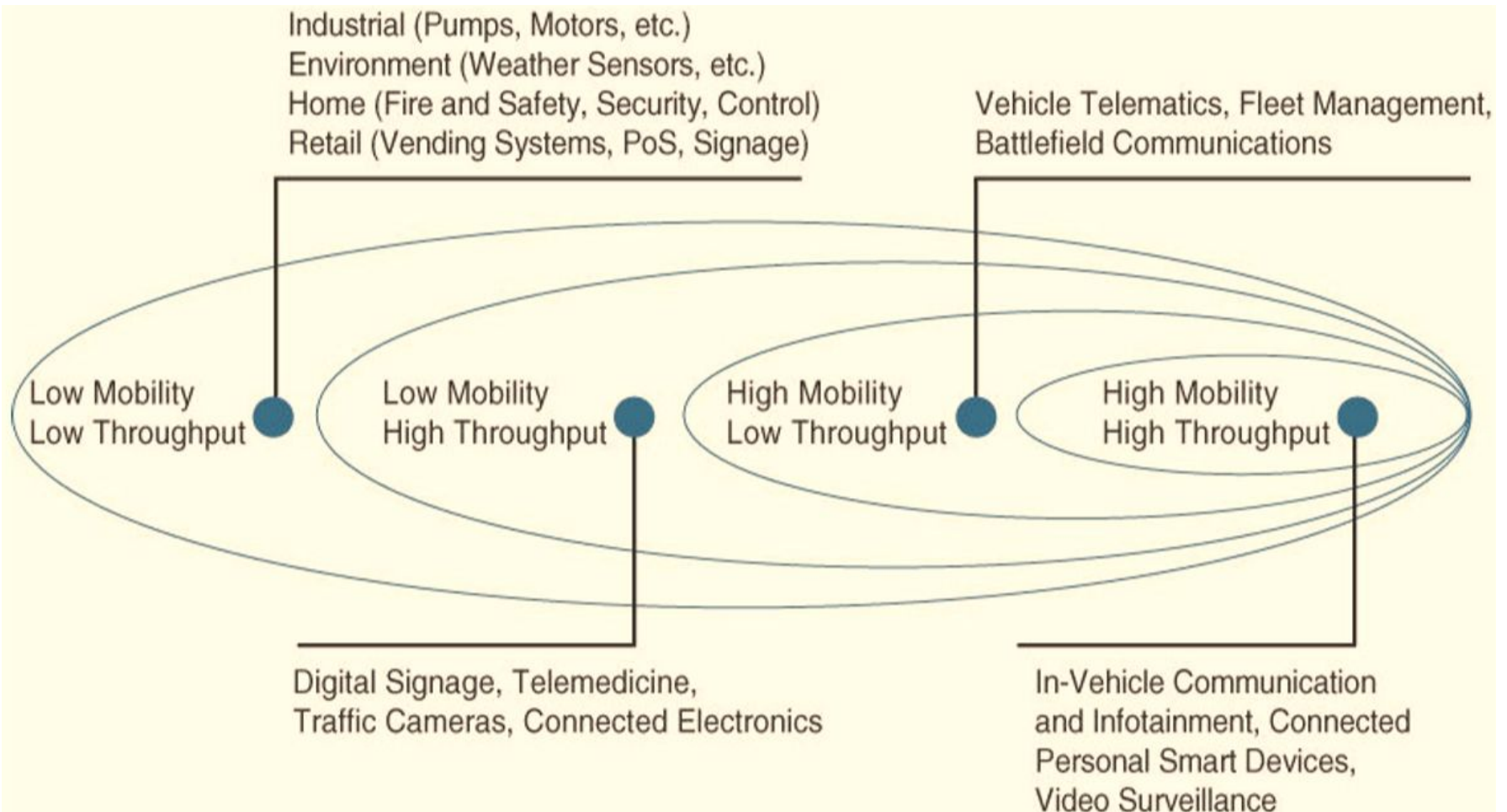


Figure 2-8 Example of Sensor Applications Based on Mobility and Throughput

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Layer 1: Things: Sensors and Actuators

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- These sensors either need to have a constant source of power (resulting in limited mobility) or need to be easily accessible for battery replacement (resulting in limited transmission range).
- A first step in designing an IoT network is to examine the requirements in terms of mobility and data transmission (how much data, how often).

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- Once you have determined the influence of the smart object form factor over its transmission capabilities (transmission range, data volume and frequency, sensor density and mobility), you are ready to connect the object and communicate.
- Compute and network assets used in IoT can be very different from those in IT environments.
- The difference in the physical form factors between devices used by IT and OT is obvious even to the most casual of observers.
 - ? What typically drives this is the physical environment in which the devices are deployed.
 - ? What may not be as inherently obvious, however, is their operational differences.
 - ? The operational differences must be understood in order to apply the correct handling to secure the target assets.

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- Temperature sensor variance
 - ? The cause for the variance is easily attributed to external weather forces and internal operating conditions.
- Humidity fluctuations can impact the long-term success of a system as well.
- In some conditions, the systems could be exposed to direct liquid contact such as may be found with outdoor wireless devices or marine condition deployments.
- Kinetic forces.
- Shock and vibration needs vary based on the deployment scenario.
- In some cases, the focus is on low amplitude but constant vibrations, as may be expected on a bushing-mounted manufacturing system.
- In other cases, it could be a sudden acceleration or deceleration

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- Solid particulates can also impact the gear.
- Hazardous location design may also cause corrosive impact to the equipment.

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- ❑ **Access Network Sublayer:**
- ❑ There is a direct relationship between the IoT network technology you choose and the type of connectivity topology this technology allows.
 - ? Each technology was designed with a certain number of use cases in mind (what to connect, where to connect, how much data to transport at what interval and over what distance).
 - ? These use cases determined the frequency band that was expected to be most suitable, the frame structure matching the expected data pattern (packet size and communication intervals), and the possible topologies that these use cases illustrate.

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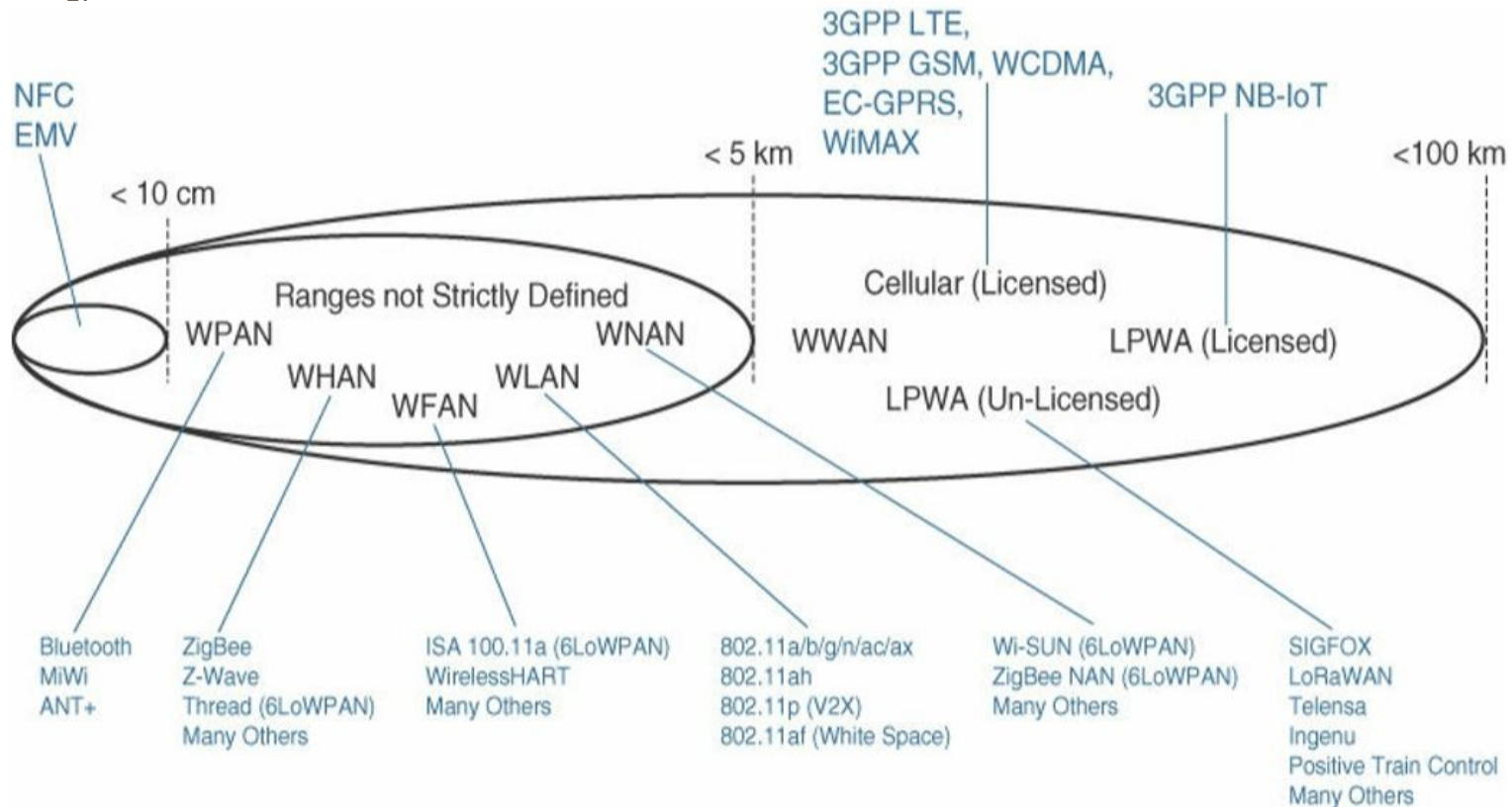
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- An access technology will be required.
 - ? IoT sometimes reuses existing access technologies
 - ? Whereas some access technologies were developed specifically for IoT use cases.

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WPAN: Wireless Personal Area Network
WHAN: Wireless Home Area Network
WFAN: Wireless Field (or Factory) Area Network
WLAN: Wireless Local Area Network

WNAN: Wireless Neighborhood Area Network
WWAN: Wireless Wide Area Network
LPWA: Low Power Wide Area

Figure 2-9 Access Technologies and Distances

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- For example,
 - ? Cellular is indicated for transmissions beyond 5 km, but you could achieve a successful cellular transmission at shorter range (for example, 100 m).
 - ? ZigBee is expected to be efficient over a range of a few tens of meters, but you would not expect a successful ZigBee transmission over a range of 10 km.

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- Common groups are as follows:
- **PAN (personal area network):** Scale of a few meters. This is the personal space around a person. A common wireless technology for this scale is Bluetooth.
- **HAN (home area network):** Scale of a few tens of meters. At this scale, common wireless technologies for IoT include ZigBee and Bluetooth Low Energy (BLE).
- **NAN (neighborhood area network):** Scale of a few hundreds of meters. The term NAN is often used to refer to a group of house units from which data is collected.

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- **FAN (field area network):** Scale of several tens of meters to several hundred meters.
 - ? FAN typically refers to an outdoor area larger than a single group of house units.
 - ? The FAN is often seen as “open space” (and therefore not secured and not controlled).
 - ? A FAN is sometimes viewed as a group of NANs,
 - FAN can be as a group of HANs or a group of smaller outdoor cells.
 - As you can see, FAN and NAN may sometimes be used interchangeably.

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- LAN (local area network):
 - ? Scale of up to 100 m.
 - In the IoT space when standard networking technologies (such as Ethernet or IEEE 802.11) are used.
 - Other networking classifications, such as MAN (metropolitan area network, with a range of up to a few kilometers) and WAN (wide area network, with a range of more than a few kilometers), are also commonly used.

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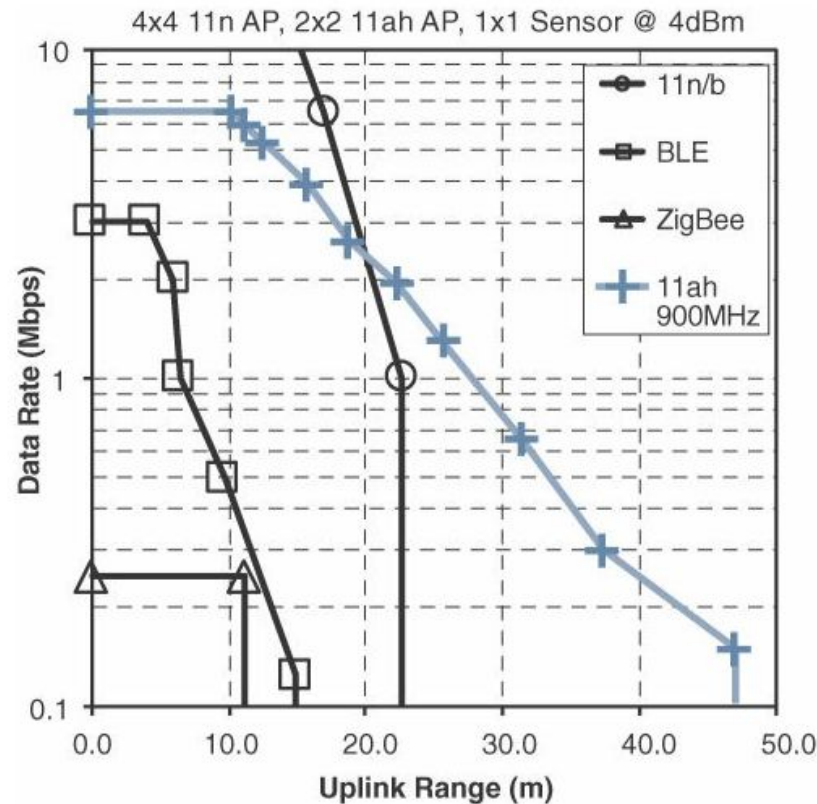
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- Similar achievable distances do not mean similar protocols and similar characteristics.
- Each protocol uses a specific frame format and transmission technique over a specific frequency (or band).
- These characteristics introduce additional differences.

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Simulation Assumptions: 1% PER, 4dB NF,
32 Bytes, D-NLOS Fading, Indoor-to-Outdoor
PL Model. 900MHz has 12dB propagation gain.

Sensor Antenna Gain: 11ah (-6.5dB)
and 11n (-4dB). AP antenna gain = 2dB.
* BT Long Range Adds 125 kbps and 500 kbps Modes

Figure 2-10 Range Versus Throughput for Four WHAN to WLAN Technologies

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- Increasing the throughput and achievable distance typically comes with an increase in power consumption.
- First, after determining the smart object requirements (in terms of mobility and data transfer),
- A second step is to determine the target quantity of objects in a single collection cell, based on the transmission range and throughput required.
 - ? This parameter in turn determines the size of the cell.

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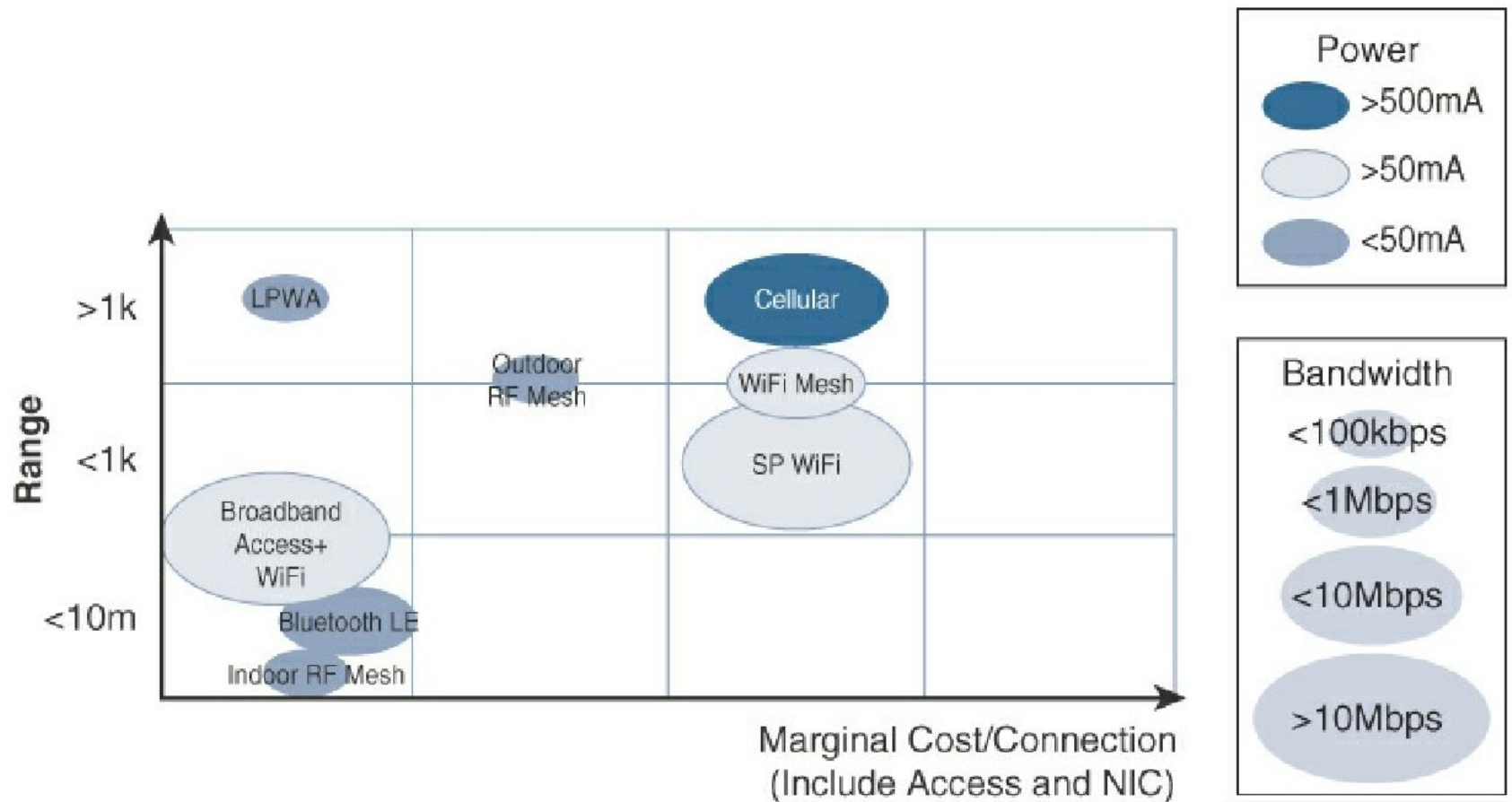


Figure 2-11 Comparison Between Common Last-Mile Technologies in Terms of Range Versus Cost, Power, and Bandwidth

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- Some technologies offer flexible connectivity structure to extend communication possibilities:
 - ? Point-to-point topologies
 - ? Point-to-multipoint topologies

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□ **Point-to-point topologies:**

- ? These topologies allow one point to communicate with another point. This topology in its strictest sense is uncommon for IoT access, as it would imply that a single object can communicate only with a single gateway.

□ **Point-to-multipoint topologies:**

- ? These topologies allow one point to communicate with more than one other point.
- ? Most IoT technologies where one or more than one gateways communicate with multiple smart objects are in this category. However, depending on the features available on each communicating mode, several subtypes need to be considered.

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- Technology that categorizes nodes based on their function is
 - ? E.g IEEE 802.15.4
- To form a network, a device needs to connect with another device. When both devices fully implement the protocol stack functions, they can form a peer-to-peer network. However, in many cases, one of the devices collects data from the others.
 - ? For example, in a house, temperature sensors

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- IEEE 802.15.4 standard,
 - ? The central point is called a coordinator for the network.
 - ? Each sensor is not intended to do anything other than communicate with the coordinator in a master/slave type of relationship.
 - ? The sensor can implement a subset of protocol functions to perform just a specialized part (communication with the coordinator).
 - Such a device is called a **reduced-function device (RFD)**.
 - An RFD cannot be a coordinator.
 - An RFD also cannot implement direct communications to another RFD.

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- The coordinator that implements the full network functions is called, by contrast, **a full function device (FFD)**.
- An FFD can communicate directly with another FFD or with more than one FFD, forming multiple peer-to-peer connections.
- Topologies where each FFD has a unique path to another FFD are called cluster tree topologies.
- FFDs in the cluster tree may have RFDs, resulting in a cluster star topology.

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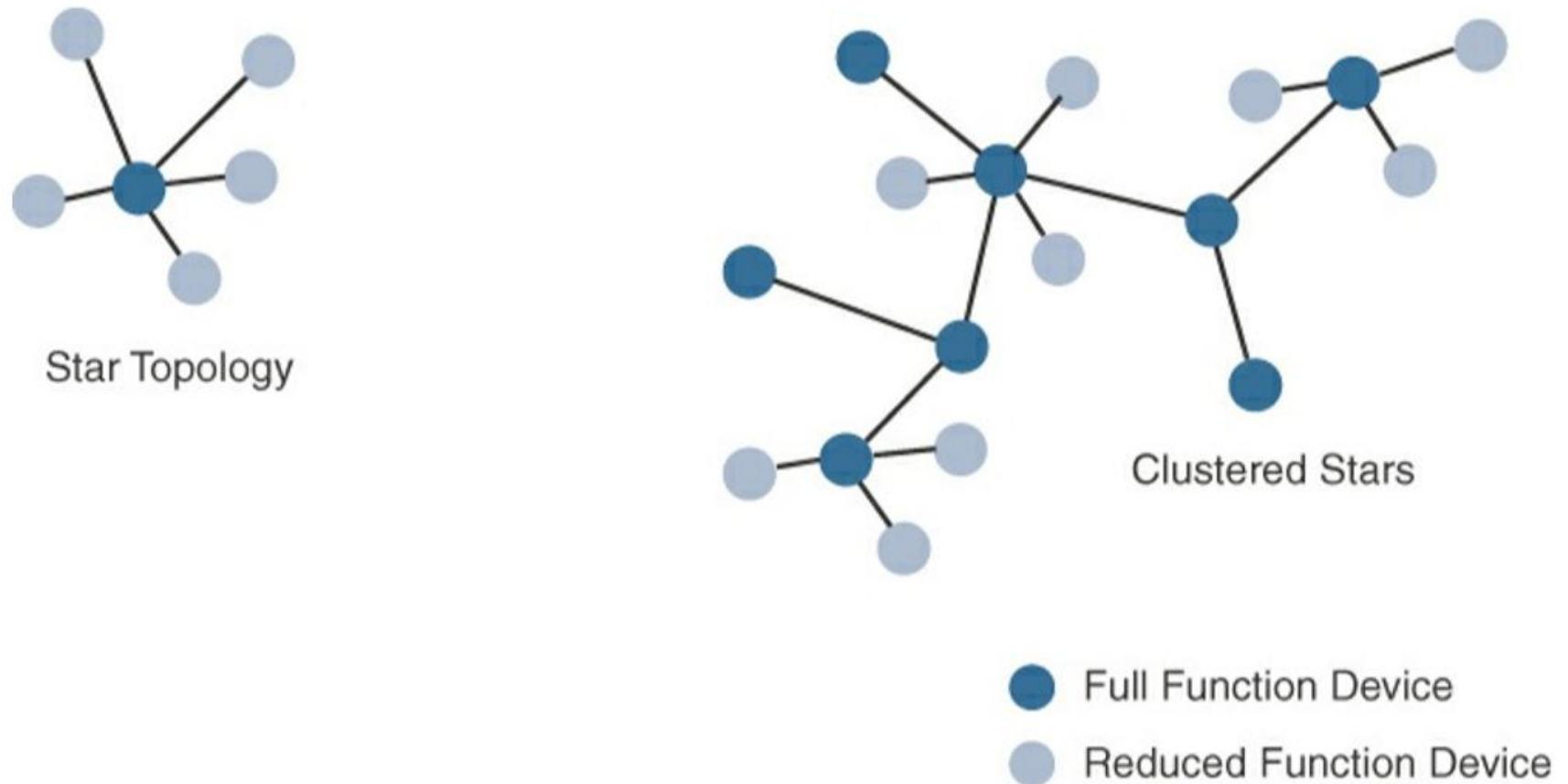


Figure 2-12 Star and Clustered Star Topologies

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□ A mesh topology

- ? Other point-to-multipoint technologies allow a node to have more than one path to another node.
- ? This redundancy means that each node can communicate with more than just one other node.
- ? This communication can extend the range of the communication link.
 - In this case, an intermediate node acts as a relay between two other nodes.
 - Range extension typically comes at the price of slower communications (as intermediate nodes need to spend time relaying other nodes' messages).
- ? An example of a technology that implements a mesh topology is Wi-Fi mesh.

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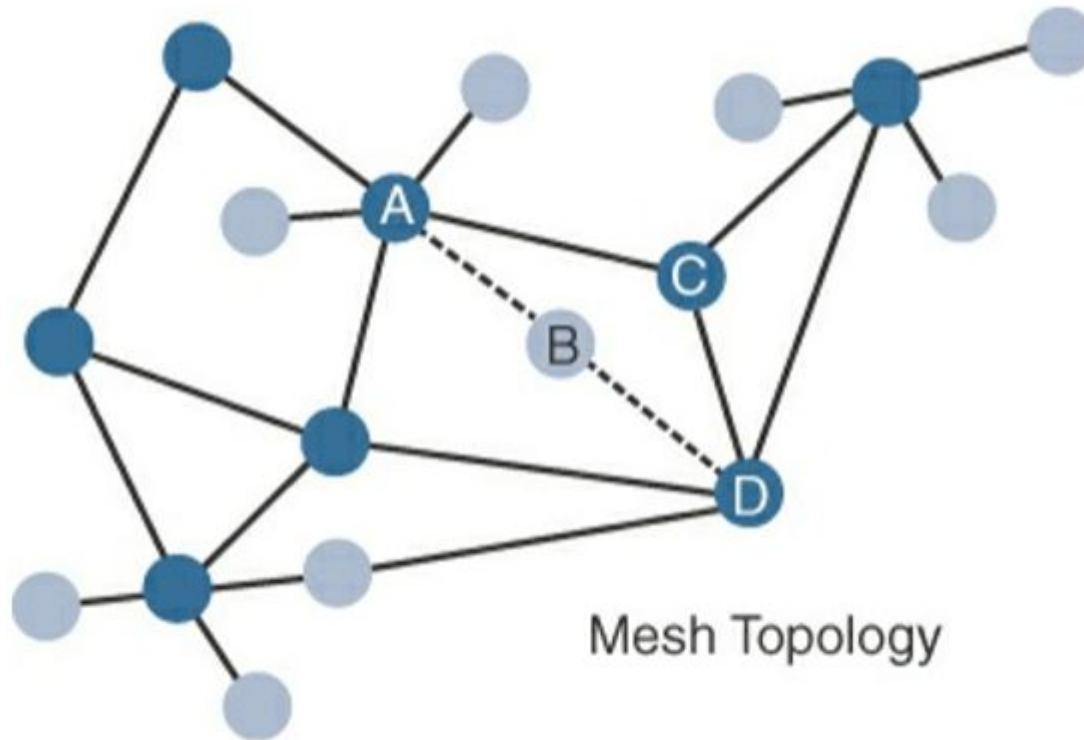


Figure 2-13 Mesh Topology

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□ Gateways and Backhaul Sublayer

- ? Data collected from a smart object may need to be forwarded to a central station where data is processed.
- ? A different location from the smart object, data directly received from the sensor through an access technology needs to be forwarded to another medium (the backhaul) and transported to the central station.
- ? The gateway is in charge of this inter-medium communication.

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- In most cases, the smart objects are static or mobile within a limited area.
- The gateway is often static.
- However, some IoT technologies do not apply this model.
 - ? For example, dedicated short-range communication (DSRC) allows vehicle-to-vehicle and vehicle-to-infrastructure communication.

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? **Dedicated short-range communication (DSRC)**

- Smart object's position relative to the **gateway is static**.
- The car includes sensors and one gateway.
- Communication between the sensors and the gateway may involve wired or wireless technologies.
- Sensors may also be integrated into the road infrastructure and connect over a wired or wireless technology to a gateway on the side of the road.
- A wireless technology (DSRC operates in the upper 5 GHz range) is used for backhaul communication, peer-to-peer, or mesh communication between vehicles.
- The range at which DSRC can communicate is limited.

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- ? For IoT architectures, the choice of a backhaul technology depends on the communication distance and also on the amount of data that needs to be forwarded.
 - E.g.
 - Environment is stable (for example, factory or oil and gas field), Ethernet can be used as a backhaul
 - Unstable or changing environments (for example, open mines) where cables cannot safely be run, a wireless technology is used E.g. WiFi
- ? **Mesh is a common topology to allow communication flexibility in this type of dynamic environment.**

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- ? However, throughput decreases as node-to-node distance increases, and it also decreases as the number of hops increases.
 - In Wi-Fi mesh network, throughput halves for each additional hop.
 - Other technologies are needed to increase beyond the range .
 - E.g. WiMax (802.16d) or cellular technology.

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Technology	Type and Range	Architectural Characteristics
Ethernet	Wired, 100 m max	Requires a cable per sensor/sensor group; adapted to static sensor position in a stable environment; range is limited; link is very reliable
Wi-Fi (2.4 GHz, 5 GHz)	Wireless, 100 m (multipoint) to a few kilometers (P2P)	Can connect multiple clients (typically fewer than 200) to a single AP; range is limited; adapted to cases where client power is not an issue (continuous power or client battery recharged easily); large bandwidth available, but interference from other systems likely; AP needs a cable
802.11ah (HaloW, Wi-Fi in sub-1 GHz)	Wireless, 1.5 km (multipoint), 10 km (P2P)	Can connect a large number of clients (up to 6000 per AP); longer range than traditional Wi-Fi; power efficient; limited bandwidth; low adoption; and cost may be an issue
WiMAX (802.16)	Wireless, several kilometers (last mile), up to 50 km (backhaul)	Can connect a large number of clients; large bandwidth available in licensed spectrum (fee-based); reduced bandwidth in license-free spectrum (interferences from other systems likely); adoption varies on location
Cellular (for example, LTE)	Wireless, several kilometers	Can connect a large number of clients; large bandwidth available; licensed spectrum (interference-free; license-based)

Table 2-4 Architectural Considerations for WiMAX and Cellular Technologies

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Layer 2: Communications Network

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□ Network Transport Sublayer

- ? In a hierarchical communication architecture in which a series of smart objects report to a gateway that conveys the reported data over another medium and up to a central station.
- ? However, practical implementations are often flexible, with multiple transversal communication paths.
 - For example, consider the case of IoT for the energy grid. Your house may have a meter that reports the energy consumption to a gateway over a wireless technology.
 - Other houses in your neighbourhood (NAN) make the same report, likely to one or several gateways.
 - If your power consumption becomes unusually high, the utility headend application server may need on-demand reporting from your meter at short intervals to follow the consumption trend.
 - From a standard vertical push model, the transport structure changes and becomes bidirectional (downstream pull model instead of upstream push).

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? **Distribution automation (DA)**

- Allows meter to communicate with neighbouring meters or other devices in the electrical distribution grid. With such communication, consumption load balancing may be optimized.
 - For example, if your air conditioning pulses fresh air at regular intervals.

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Layer 2: Communications Network Layer

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- ? Your smart meter may communicate with your house appliances to evaluate their type and energy demand.
- ? Once the system learns your consumption pattern,
- ? Smart meter may communicate with your house appliances to evaluate their type and energy demand.
 - system learns your consumption pattern
 - E.g. public car charging station

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- ? Communication structure may include
 - peer-to-peer (for example, meter to meter), point-to-point (meter to headend station), point-to-multipoint (gateway or head end to multiple meters), unicast and multicast communications (software update to one or multiple systems).
 - In a multitenant environment (for example, electricity and gas consumption management), different systems may use the same communication pathways.
 - This communication occurs over multiple media (for example, power lines inside your house or a short-range wireless system like indoor Wi-Fi and/or ZigBee), a longer-range wireless system to the gateway, and yet another wireless or wired medium for backhaul transmission.
- ? Communication structure must include, a network protocol with specific characteristics, Scalability, security, Standard and open protocol, IP protocol .

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□ **Transport layer protocols**

- ? built above IP (UDP and TCP) can easily be leveraged to decide whether the network should control the data packet delivery (with TCP) or whether the control task should be left to the application (UDP).
- ? UDP is a much lighter and faster protocol than TCP.
 - However, it does not guarantee packet delivery.
- ? Both TCP and UDP can be secured with TLS/SSL (TCP) or DTLS (UDP).

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- **IoT Network Management Sublayer**
 - ? IP, TCP, and UDP bring connectivity to IoT networks.
 - ? Upper-layer protocols need to take care of data transmission between the smart objects and other systems.
 - ? Multiple protocols have been leveraged or created to solve IoT data communication problems.
 - ? Some networks rely on a
 - **push model**
 - that is, a sensor reports at a regular interval or based on a local trigger
 - **pull model**
 - that is, an application queries the sensor over the network
 - **Multiple hybrid approaches** are also possible.

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- **Web-based protocols**
 - ? **HTTP**
 - ? **WebSocket**
 - ? **Extensible Messaging and Presence Protocol (XMPP)**

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- Some IoT implementers have suggested **HTTP** for the data transfer phase.
 - ? HTTP has a client and server component.
 - The sensor could use the client part to establish a connection to the IoT central application (the server),
 - but HTTP is something of a fat protocol and was not designed to operate in constrained environments with low memory, low power, low bandwidth, and a high rate of packet failure.

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- ❑ **WebSocket** (Small compared to HTTP)
 - ? is part of the HTML5 specification, and provides a simple bidirectional connection over a single connection.
 - ? WebSocket to manage the connection between the smart object and an external application.
 - ? WebSocket is often combined with other protocols, such as MQTT (Message Queue Telemetry Transport).

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- **Extensible Messaging and Presence Protocol (XMPP)**
 - ? XMPP is based on instant messaging and presence.
 - ? It allows the exchange of data between two or more systems and supports presence and contact list maintenance.
 - ? It can also handle publish/subscribe, making it a good choice for distribution of information to multiple devices.
 - ? A limitation of XMPP is its reliance on TCP,
 - which may force subscribers to maintain open sessions to other systems and may be a limitation for memory-constrained objects.

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- **IoT protocol**
 - ? **IETF Constrained Restful Environments (CoRE)**
 - ? **Constrained Application Protocol (CoAP).**
 - ? **MQTT**

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- ❑ **IETF Constrained Restful Environments (CoRE)**
- ❑ Working group: **Constrained Application Protocol (CoAP).**
 - ? CoAP uses some methods similar to those of HTTP (such as Get, Post, Put, and Delete) but implements a shorter list, thus limiting the size of the header.
 - ? CoAP also runs on UDP (whereas HTTP typically uses TCP).
 - ? CoAP observation: Observation allows the streaming of state changes as they occur,

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- ❑ **Message Queue Telemetry Transport (MQTT)**
- ❑ MQTT uses a broker-based architecture.
 - The sensor can be set to be an **MQTT publisher** (publishes a piece of information),
 - the application that needs to receive the information can be set as the **MQTT subscriber**,
 - Any intermediary system can be set as a **broker** to relay the information between the publisher and the subscriber(s).
- ? MQTT runs over TCP.
- ? A consequence of the reliance on TCP is that an MQTT client typically holds a connection open to the broker at all times.
- ? This may be a limiting factor in environments where loss is high or where computing resources are limited.

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Layer 3: Applications and Analytics

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- Once connected to a network, your smart objects exchange information with other systems.

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- **Analytics Versus Control Applications**

- ? Each application collects data and provides a range of functions based on analysing the collected data.

- **Analytics application:**

- ? This type of application collects data from multiple smart objects, processes the collected data, and displays information resulting from the data that was processed.

- **Control application:**

- ? This type of application controls the behavior of the smart object or the behavior of an object related to the smart object.
 - For example, a pressure sensor may be connected to a pump. A control application increases the pump speed when the connected sensor detects a drop in pressure.

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- **Data Versus Network Analytics**

- **Data analytics:**

- ? This type of analytics processes the data collected by smart objects and combines it to provide an intelligent view related to the IoT system.

- ? E.g.

- Simple case dashboard can display an alarm when a weight sensor detects that a shelf is empty in a store.
 - Complex case, temperature, pressure, wind, humidity, and light levels collected from thousands of sensors
 - Data analytics can also monitor the IoT system itself.
 - For example, a machine or robot in a factory can report data about its own movements.

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❑ **Network analytics:**

- ? Most IoT systems are built around smart objects connected to the network.
- ? A loss or degradation in connectivity is likely to affect the efficiency of the system.
- ? Such a loss can have dramatic effects.
 - For example, open mines use wireless networks to automatically pilot dump trucks.
- ? Loss of connectivity means that data stops being fed to your data analytics platform, and the system stops making intelligent analyses of the IoT system.

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■ **Data Analytics Versus Business Benefits**

- ? Collecting and interpreting the data generated by these devices is where the value of IoT is realized.
- ? Static IoT networks : Cant be update or add/remove sensors
- ? Connect to new sensors.
 - This enhanced data processing and can add value for businesses.
- ? E.g. Vending machines deployed throughout a city
 - Sensors can be deployed to report when a machine is in an error state.
 - A repair person can be sent to address.
 - This type of alert is a time saver

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- ❑ **Smart Services:**
- ❑ Smart services use IoT and aim for efficiency.
 - ? Smart services can also be used to measure the efficiency of machines by detecting machine output, speed, or other forms of usage evaluation.
- ❑ Entire operations can be optimized with IoT.
- ❑ Smart services can be integrated into an IoT system.
 - ? For example, sensors can be integrated in a light bulb.
- ❑ Similar efficiency can be extended to larger systems than a house.
 - ? For example, smart grid